

NUVOTON MS51FB9AE MICROCONTROLLER ARCHITECTURE

- CORE:1T 8051 8-bit microcontroller.
- Maximum clock speed is upto 24 MHz.
- 16KB flash memory.
- 256 bytes internal memory and 1KB SRAM.
- 2.5v to 5.5 operating voltage.
- In CPU we have ALU ,Accumulator, B registers, R0-R7 general purpose working registers, program counter, stack pointer, data pointer, program status word .
- The memory is divided into two ,Program memory :APROM and LDRAM, Data memory: internal RAM and XRAM.
- It as registers like Accumulator ,b registers,R0 to R7 general purpose working registers, data pointers ,program counters, stack pointers, program status word.
- We have instruction set:
Data transfer: MOV, MOVX, MOVC.
Arithmetic: ADD, SUB, INC, DEC, MUL, DIV.
Logical :ANL, ORL, XRL, CPL.
Branch: SJMP,LJMP ,CALL, RET ,DJNZ .
Bit operation: SETB, CLR,JB,JNB.
- Who the process work?
PC→fetch→decode→execute→update registers→update PC→next instruction.
- Peripherals includes GPIO, timer 0/1,Timer2,Timer3,PWM,Watch Dog Timer,UART,SPI,I²C,12 Bit ADC, Self Wake Up Timer.